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REPORT

Experimental determination of the cost of lesion healing on *Porites compressa* growth

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Abstract The spotted puffer (*Arothron meleagris*) commonly feeds on finger coral (*Porites compressa*) on Hawaiian shallow reefs. The predation involves removal of branch tips, resulting in open lesions that, in the field, heal in little over a month. This study determined the cost of tissue regeneration across simulated bite scars on the growth of experimentally scarred *P. compressa* nubbins, of three different size classes (4, 2, 1 cm tall), in a controlled laboratory environment. The results indicate that inflicted *P. compressa* lesions heal completely within approximately 36 days, on even the smallest nubbins. While growth was slowed before lesions closed, the healing process did not appear to affect the longer-term growth of fragments. Successful regeneration of tissue may be explained by small lesion size, the high surface area to perimeter ratio of the lesions, and the location of lesions on the growing tips. The lack of impact on overall growth rate of experimental fragments indicates that under favorable conditions, *P. compressa* is resilient to predation.

Keywords Lesion · Healing · Regeneration · Predation · Growth · *Porties compressa*

Introduction

Feeding by the spotted puffer *Arothron meleagris* (Lacepède 1798) on the tips of the finger coral *Porites compressa* (Link 1807) is common on Hawaiian reefs (Jayewardene et al. 2009) and involves removal of tissue and chunks of coral skeleton. Despite this damage, field observations indicate that *P. compressa* bite lesions heal successfully in a little over 40 days (Jayewardene et al. 2009).

Tissue regeneration across coral lesions is primarily fueled by polyps and tissue directly bordering the lesions (Meesters et al. 1994, 1997; Oren et al. 1997). The rate of regeneration is dependent on the length of the living tissue bordering the lesion and the lesion surface area to perimeter ratio (Meesters et al. 1994, 1997; Oren et al. 1997; van Woesik 1998; Lirman 2000). Regeneration also depends on the coral species (Bak 1983; Meesters et al. 1992), coral morphology (Meesters et al. 1996; Hall 1997), reproductive state (Kramarsky-Winter and Loya 2000), and environmental conditions (Lester and Bak 1985; Titlyanov et al. 2005). Allocation of energy to tissue regeneration adversely affects coral growth (Bak 1983; Meesters et al. 1994), fecundity (Rinkevich and Loya 1989; Van Veghel and Bak 1994), disease resistance, and competitive ability (Bak and Criens 1981). As the quantity of tissue fueling regeneration is finite, regeneration may stop at some point, in favor of other life functions, such as growth, causing unhealed portions of lesions to become permanent (Meesters et al. 1994). Algae and invertebrates may invade the bare substrate (Titlyanov et al. 2005), leading to competition and possible coral mortality (Meesters et al. 1996, 1997). In the face of frequent predation and increasing anthropogenic influences, continuous allocation of energy to healing may compromise the function and survival of corals (Meesters and Bak 1993; Meesters et al. 1992).

The specific objective of this study was to compare the growth rates of experimentally scarred *P. compressa* nubbins of three different sizes classes (4, 2, 1 cm tall) with undamaged nubbins in a controlled laboratory

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environment. Nubbin growth was measured during the tissue regeneration-healing process and after lesion closure using the buoyant weight method (Jokiel et al. 1978).

Materials and methods

Experimental design and preparation

Sixty-four *Porites compressa* nubbins were haphazardly collected from the reef at approximately 2 m depth at the Hawaii Institute of Marine Biology (HIMB). Of these specimens, 32 randomly chosen nubbins were cut to 5 cm lengths, 16 nubbins were cut to 3 cm lengths, and another 16 nubbins were cut to 2 cm lengths. Using epoxy putty, each nubbin was individually attached to a 2.5 cm × 2.5 cm ceramic tile. Sixteen of the 5-cm nubbins were allocated as controls and remained undamaged, while *Arothron meleagris* bites were simulated on the remainder of the nubbins by removing ~1 cm height off each of the nubbin tips using a 7-inch wire cutter (Fig. 1). This created four experimental treatment groups (5-cm undamaged controls, 4-cm scarred nubbins, 2-cm scarred nubbins, and 1-cm scarred nubbins), which were placed randomly in relation to each other in a 1 m (w) × 1 m (l) × 0.4 m (h) body of water in a mesocosm tank at HIMB. The nubbins in the tank were shaded from direct irradiance by a mesh and subject to a continuous flow (7–8 l min^{-1}) of sea water, pumped from the adjacent reef. The sea water was aerated and exposed to ambient air temperature. As closure of lesions was expected to occur within 42 days, based on previous field observations (Jayewardene et al. 2009), nubbins were monitored for a total of 3 months (April 7, 2007–June 28, 2007) to enable measurement of growth both leading up to the closure of lesions, and for an extended period thereafter. Average daily air temperature and average daily solar radiation measured during the experimental time period at the HIMB weather station was 25.7 ± 1.1°C (mean ± SD), respectively, 252.9 ± 45.1 calories cm^{-2} (mean ± SD). Average daily sea water temperature, at the sea water intake (2 m depth on adjacent reef), was 24.4 ± 0.6°C (mean ± SD).

Fig. 1 The four treatment groups: 5-cm undamaged controls, 4-cm tall scarred nubbins, 2-cm tall scarred nubbins, and 1-cm tall scarred nubbins

Measurement of healing and growth

Growth following scarring (=day 0) was measured on day 1, 3, 5, 7, 10, 12, 14, 18, 20, 25, 31, 34, 41, 48, 56, 62, 66, 75, 80, and 83. The stage of tissue regeneration across lesions was visually assessed and photographically recorded at each of these times, with the aim of determining the average number of days until lesion closure. The buoyant weight method (Jokiel et al. 1978) was used to measure skeletal growth. This involved removing each nubbin from the mesocosm tank, hanging it in a beaker of salt water from a balance, and recording the weight in grams to three decimal places.

To compare growth rates between treatment groups, the weight gained per unit area of live tissue (g cm^{-2}) was calculated for each nubbin. The surface area of live tissue for each nubbin was calculated upon completion of monitoring by: (1) weighing each nubbin, dipping the nubbin excluding the bottom surface into hot wax (60°C), letting the wax cool, and reweighing the nubbin with the wax coat on; (2) similarly, weighing and dipping into the wax 12 standard calcium carbonate blocks with varying yet known surface areas, reweighing each of these when covered with wax, calculating the weight of wax per surface area for each block, then calculating the mean weight of wax per carbonate surface area (36.9 ± 6.6 g cm^{-2} SD); (3) using this standard weight-surface ratio to calculate the total surface areas of each of the nubbins based on the weights of their wax coats (Stimson and Kinzie 1991). Nubbin tissue surface area for the control was 28.1 ± 4.2 cm^2 (mean ± SD), 21.4 ± 4.2 cm^2 (mean ± SD), 8.9 ± 1.2 cm^2 (mean ± SD), and 5.0 ± 1.0 cm^2 (mean ± SD) for the 4-, 2-, and 1-cm treatment, respectively. The surface area of each inflicted lesion was calculated by measuring the two perpendicular diameters [0.9 ± 0.3 cm^2 (mean ± SD) (same for all treatments, one-way ANOVA: $f = 0.77$, $P = 0.469$)]. The entire surface area of the lesion was subtracted from the calculated nubbin surface area, when calculating weight gain per nubbin surface area on day 0. Subsequently, proportionately less of the lesion area was subtracted from the nubbin surface area until nothing was subtracted on the day of lesion closure. The relevant proportions of lesion area subtracted for each nubbin were calculated by generating a set of representative linear equations using the range in number of healing days and the assumption that healing was 0% on day 0 and 100% on the day of lesion closure.

Data analysis

Lesion-healing rates were compared among treatments using a one-way ANOVA. Nubbin growth rates among

treatments and before and after lesion healing were compared using factor analysis (General Linear Model).

Results

Healing and growth

Complete lesions closure occurred within 39 ± 2 days (mean ± SE) for 4-cm treatment, 35 ± 1 days (mean ± SE) for 2-cm treatment, and 34 ± 2 days (mean ± SE) for 1-cm treatment. These healing rates were not statistically different (one-way ANOVA: $f = 2.47$, $P = 0.096$). Apart from an initial loss in weight of 0.001 ± 0.0004 g cm^{-2} (mean ± SE) during the first 4 days after scarring for the smallest treatment (1 cm), average cumulative weight gain (g cm^{-2}), for each of all four treatments over the course of the 3-month monitoring period, was positive and linear (Fig. 2). The factor analysis indicated that the rate of nubbin growth [(end weight–start weight) × 30 days] was significantly different both between treatments and before versus after lesion closure (period) (Table 1). The interaction term (treatment × period) was also significant, indicating that the difference in growth rates among treatments varied between periods (Fig. 3). There was no significant difference in growth rate among size classes before lesion closure (one-way ANOVA: $f = 2.52$, $P = 0.067$), but there was a significant difference after lesion closure (one-way ANOVA: $f = 8.13$, $P < 0.001$). A Tukey's multiple comparisons test indicated that the difference after lesion closure was between the control versus 1-cm treatment, and between the 4 versus 2-cm and 1-cm treatments. The difference in growth rate before versus after was significant for the control and 4-cm treatment (paired comparisons *t*-test: $P = 0.05$, respectively, $P = 0.05$), and clearly significant for the 2-cm and the 1-cm treatment (paired comparisons *t*-test: $P < 0.001$, respectively, $P < 0.001$) (Fig. 3).

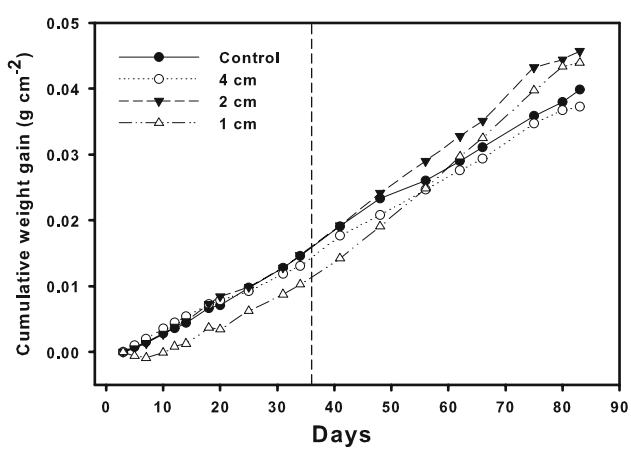

Fig. 2 The average cumulative weight gain (g cm^{-2}) for 5-cm undamaged controls, 4-cm treatment nubbins, 2-cm treatment nubbins, and 1-cm treatment nubbins over the course of the 3-month monitoring period. The *dashed line* indicates the average time of scar healing

Table 1 Factorial analysis using general linear model to test differences in mean nubbins growth rates between the controls and three scarred nubbins size classes (=treatments), before and after scar healing (=period)

Factor	*df*	MS	*F*	*P*
Treatment	3	0.0000604	2.74	0.047
Period	1	0.0010294	46.64	<0.001
Treatment × period	3	0.0001832	8.30	<0.001
Error	120	0.0000221		

Treatment and period were both fixed factors

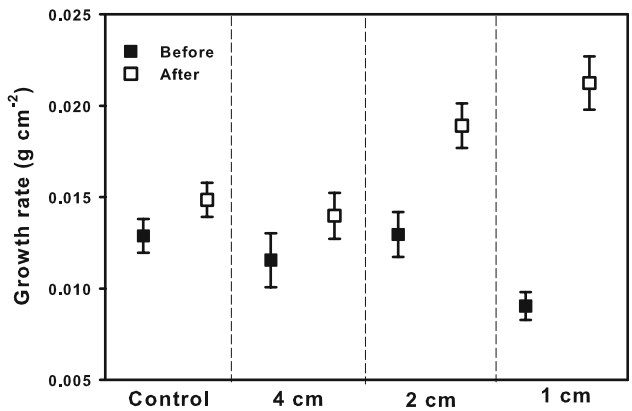

Fig. 3 The growth rates before and after healing for 5-cm undamaged controls, 4-cm treatment nubbins, 2-cm treatment nubbins, and 1-cm treatment nubbins. *Error bars* indicate ±1 SE

Discussion

This study indicates that experimental lesions, simulating *A. meleagris* bites, on *P. compressa* heal in a little over a month, and with little effect on the long-term growth of even very small coral fragments (1 cm).

Fast and complete regeneration of tissue across *P. compressa* is likely because of the small size of the bite lesions. *A. meleagris* feeds on the branch tips of the finger-like *P. compressa*, and therefore inflicted lesions will tend to be circular in shape and similarly limited in size (to the size of coral branches). Rapid healing of lesions may also be explained by polyps at the tip of branches being younger and more active than polyps at the base, as documented for branching *Acropora palmata* in the Caribbean (Meesters and Bak 1995). Furthermore, as branched corals are more prone to fragmentation, they tend to have rapid rates of tissue regeneration (Hall 1997), which promotes successful

fusion to new substrate (Bak 1983). The lack of effect of *P. compressa* nubbin size on tissue healing has also been documented for *A. palmata* (Lirman 2000) and may be explained by lesions on even the smallest nubbins being surrounded by the minimum amount of tissue necessary to fuel regeneration (Meesters et al. 1994, 1997; Oren et al. 1997).

While the initial brief weight loss in the smallest *P. compressa* size class indicates a negative response to lesion infliction, the ability of these very small fragments to recover and continue growth throughout and beyond the lesion-healing process indicates that predation does not constitute a major energy drain on *P. compressa*. This is possibly because environmental conditions under which the fragments were maintained in the laboratory were favorable (non-extreme temperatures, protection from irradiance, continuous sea water flow and aeration), potentially resulting in a reserve of energy available for tissue regeneration (Muscatine et al. 1984; Spencer Davies 1984; Edmunds and Spencer Davies 1984; Nagelkerken et al. 1999).

While growth rates were similar among nubbin size classes, during lesion healing, growth rates were enhanced after lesion closure, especially for the smallest nubbins (1 and 2 cm). Hypothetically, the lack of suppression of growth rates during healing might have been because of extra resources available for tissue regeneration. The period of enhanced growth following lesion closure, in the smaller nubbins, may have been because smaller nubbins required disproportionally more resources to heal compared with large nubbins. Growth rates for the smaller nubbins did appear to level off toward the end of the experimental period, indicating a return to baseline growth rates. Growth rates were also somewhat higher in the latter half of the experiment for the larger nubbins, possibly a result of stimulated growth (Jokiel and Coles 1977; Lester and Bak 1985; Miller 1995; Titlyanov et al. 2005) from increasing sea water temperature during the course of the experiment. The surface temperature of the sea water, feeding the experimental tank, was 23.8 ± 0.3°C (mean ± SD) in April, compared to 24.3 ± 0.4°C (mean ± SD) and 25.1 ± 0.2°C (mean ± SD) in May and June, respectively.

The experimental findings indicate that there is little predation cost to *P. compressa* growth, unless environmental conditions interfere that are less than optimal for growth, and/or *P. compressa* is exposed to stressors such as disease, bleaching, and sedimentation, which have been shown to slow the healing of lesions in other corals (Bak and Criens 1981; Meesters et al. 1992; Meesters and Bak 1993; Kramarsky-Winter and Loya 2000; Fine et al. 2002; Fisher et al. 2007). Furthermore, growth might be suppressed by continuous allocation of energy toward increased frequency of lesion generation resulting from intensified predation. This might also occur if predation pressure remains the same, while there is a reduction in the population abundance of *P. compressa* (Jayewardene et al. 2009), a decrease in the length of branches and/or a higher occurrence of small fragments due to fragmentation.

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